Homework — 1.1.1 Structure and Function of the Processor

Name: ____ Date: _ ***Mark:*** ___ / 20

OCR H446 — set after the 1.1.1 lesson. Show your working.

Part A — This week's topic (~14 marks)

1. State the **full name** and **role** of each of the following registers. **[3]** *(AO1)*

- PC: ______
- ACC: ______
- MAR: ______

2. Describe the role of the **Control Unit (CU)** within the processor. Give **two** distinct points. **[2]** *(AO1)*

3. The **fetch** stage of the fetch–decode–execute cycle copies the contents of the PC to the MAR. State the **three** further steps that complete the fetch stage, in the correct order. **[3]** *(AO1)*

4. A processor has a **data bus** that is 32 bits wide and an **address bus** that is 20 bits wide.

(a) Calculate the maximum number of memory locations that can be addressed. Show your working. **[2]** *(AO2)*

(b) State **one** effect of widening the **data bus** on the performance of the processor. **[1]** *(AO1)*

5. Explain how adding more **cores** to a processor can improve performance, and state the **condition** that must be met for the benefit to be realised. **[2]** *(AO1)*

6. During the **decode** stage, the Control Unit splits the contents of the CIR into two parts. Name both parts and state what each one represents. **[2]** *(AO1)*

Part B — Retrieval: keep it fresh (~6 marks)

7. State the direction (**unidirectional** or **bidirectional**) and the contents carried by each of the following buses. **[3]**

- Address bus: ______
- Data bus: ______
- Control bus: ______

8. Explain what **cache** memory is and **one** reason it improves CPU performance. **[2]**

9. State which cache level (**L1** or **L3**) is normally **larger**. **[1]**